Animal-Image-Classifier

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1. Introduction and Project Overview

The *Animal-Image-Classifier* project focuses on developing a deep learning system capable of classifying animal images into five categories: dog, cow, cat, lamb, and zebra. The project, described in the report *Multiclass Animal Image Classification: CNN and YOLOv11 Based Deep Learning Approach*, compares two image classification approaches: a custom Convolutional Neural Network (CNN) trained from scratch and a pretrained YOLOv11 classification model adapted through transfer learning.

My contribution to the project concerns the creation and organization of the image dataset and its preparation for training and evaluating the classification models. The dataset serves as the foundation for teaching the models to distinguish between the five animal categories. Its organization, labeling, preprocessing, and appropriate use are essential for producing reliable classification results.

2. Dataset Creation and Organization

The dataset contains 500 images, with 100 images assigned to each animal category. The images were collected from publicly available online sources and smartphone photographs. This combination was intended to introduce variations in image resolution, lighting, animal poses, and background complexity.

The five categories are:

- **Dog:** Images representing dogs under different visual conditions.
- **Cow:** Images showing cows with variations in appearance and background.
- **Cat:** Images representing cats in different poses and settings.
- **Lamb:** Images of lambs with different visual characteristics.
- **Zebra:** Images showing zebras, including their distinctive striped patterns.

Each image must be associated with the correct class label so that the models can learn the relationship between visual features and animal categories. Maintaining a balanced distribution of 100 images per class helps prevent the dataset from favoring a particular category because of its image count.

My dataset-related contribution is important because a well-organized collection of labeled images makes the training process more systematic and supports a meaningful comparison between the two models. However, the available report does not identify individual contributions

or provide a detailed record of which images were personally collected, labeled, or verified by each team member.

3. Dataset Preprocessing

Before training, the images undergo preprocessing to make them compatible with the deep learning models. The project report identifies resizing, pixel normalization, and data augmentation as important preparation steps.

First, images collected from different sources may have different dimensions and resolutions. Resizing creates consistent input dimensions. The custom CNN uses RGB images with dimensions of 256 × 256 × 3, representing height, width, and three color channels.

Second, normalization standardizes pixel intensities. A common approach for eight-bit images is to divide each pixel value by 255, converting values from the range 0–255 to 0–1. The report confirms normalization but does not specify the exact formula used in the implementation.

Third, data augmentation introduces additional visual variation. The reported techniques include horizontal flipping, random rotation, scaling, and zooming. These transformations help reduce overfitting by exposing the CNN to different versions of training images.

The dataset is divided into training, validation, and test subsets using a 70:20:10 ratio. For 500 images, this corresponds to 350 training images, 100 validation images, and 50 test images. The training set is used to learn model parameters, the validation set helps monitor training and tune settings, and the test set is reserved for final evaluation.

4. Dataset Use in CNN Training

The CNN is implemented using Python and TensorFlow/Keras. Its architecture contains three convolutional layers with 16, 32, and 64 filters, followed by max-pooling operations. These layers progressively extract visual features, from simple edges and textures to more complex patterns associated with animal appearances.

The extracted feature maps are flattened and passed to a dense layer containing 128 neurons. A dropout rate of 0.5 is used to help reduce overfitting. The final layer contains five neurons with softmax activation, producing class probabilities for dog, cow, cat, lamb, and zebra.

During training, the CNN processes batches of labeled images and produces predictions. Categorical cross-entropy measures the difference between the predicted probabilities and the correct labels. The Adam optimizer uses the resulting gradients to update the model's weights. This process repeats across training epochs, while validation loss is monitored for early stopping.

The same dataset also supports the YOLOv11 classification model. Unlike the CNN trained from scratch, YOLOv11 uses pretrained weights and transfer learning to adapt previously learned

visual representations to the five animal classes. This approach can reduce training time when the available dataset is small.

5. Results and Conclusion

The project report states that the CNN achieved 100% test accuracy and YOLOv11 achieved 94%. It also reports faster convergence for YOLOv11, while the CNN provides a simpler architecture. These results illustrate how the same dataset can be used to compare models trained through different learning strategies.

The dataset's contribution extends beyond supplying images: its class labels, organization, preprocessing, and separation into training, validation, and test sets establish the foundation for the entire experiment. This work demonstrates the importance of dataset preparation in computer vision and provides practical experience with image preprocessing, CNN training, transfer learning, and model evaluation.

Nevertheless, the dataset contains only 500 images, so additional independent testing is necessary before concluding that either model will generalize reliably to diverse real-world animal images. The reported results should also be interpreted cautiously because the project report contains an inconsistent statement suggesting accuracy above 100%, which is not possible under the standard accuracy definition.